

Program : Diploma in Robotic Process Automation	
Course Code : 5302	Course Title: Data Mining
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To understand data warehouse concepts, architecture, business analysis and tools
- To understand data pre-processing and data visualization techniques
- To study algorithms for finding hidden and interesting patterns in data
- To understand and apply various classification and clustering techniques using tools.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principles and theorems of Engineering Mathematics	1002 & 2002	Mathematics I &II	1 & 2
Basic knowledge of Database	3133	Database Management System	3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the core concepts of Data Mining and Data Warehousing	12	Understanding
CO2	Apply appropriate techniques to convert raw data into suitable format for practical data mining tasks and understand different Classification Models.	18	Applying

CO3	Analyze and compare various classification algorithms and apply in appropriate domain.	15	Applying
CO4	Make use of the concept of association rule mining in real world scenario. Select appropriate clustering and algorithms for various applications	13	Applying
	Series Test	2	

Course Outcomes:

On completion of the course student will be able to:

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	1				2
CO2	3	2	2				2
CO3	3	2	2				2
CO4	3	2	2				2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the core concepts of Data Mining and Data Warehousing		
M1.01	Data Mining Concepts and Applications	1	Understanding
M1.02	Data Mining Stages	1	Understanding
M1.03	Data Mining Models	1	Understanding
M1.04	Applications of Data Mining	1	Understanding

M1.05	Data warehouse and its architecture	2	Understanding
M1.06	OLAP vs OLTP	1	Understanding
M1.07	OLAP operations	2	Understanding
M1.08	Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation,	2	Understanding
M1.09	Applications of DWH	1	Understanding

Contents:

Introduction to Data Mining – Data Mining Concepts and Applications, Data Mining Stages, Data Mining Models, Applications of Data Mining, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), OLAP vs OLTP, OLAP operations, Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation, Applications of DWH .

CO2:	Understanding and Apply various Data Preprocessing steps and Classification Models		
M2.01	Data Preprocessing Concepts	1	Understanding.
M2.02	Data Cleaning	1	Applying
M2.03	Data Integration	1	Applying
M2.04	Data Reduction Strategies Overview-Dimensionality Reduction, Numerosity Reduction	2	Understanding.
M2.05	Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation	2	Understanding.
M2.06	Data Transformation Strategies Overview - Normalization- min-max normalization, z-score normalization	2	Applying
M2.07	Data Transformation Strategies Overview - Discretization, Concept Hierarchy Generation	2	Understanding
M2.08	Introduction to Classification and Prediction, Issues regarding classification and prediction	1	Understanding
M2.09	Decision Tree- ID3	3	Applying
M2.10	Naive Bayes Classifier.	3	Applying
	Series Test – I	1	

Contents:

Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction Strategies-Dimensionality Reduction, Numerosity Reduction, Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation, Normalization- min-max normalization, z-score normalization, Discretization, Concept Hierarchy Generation

Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction, Decision Tree- ID3, Naive Bayes Classifier.

CO3:	Analyze and compare various classification algorithms and apply in appropriate domain.		
M3.01	Artificial Neural Networks- Concepts and characteristics	2	Understanding
M3.02	Back propagation Algorithm.	3	Applying
M3.03	Support Vector Machines	2	Understanding.
M3.04	Lazy Learners-K Nearest Neighbor Classifier.	3	Applying
M3.05	Accuracy and error Measuresevaluation.	2	Applying
M3.06	Prediction:-Linear Regression	3	Applying

Contents:

Artificial Neural Network-concepts and Characteristics, Back propagation Algorithm. Support Vector Machines, Lazy Learners-K Nearest Neighbor Classifier. Accuracy and error Measures evaluation. Prediction:-Linear Regression

CO4:	Make use of the concept of association rule mining in real world scenario. Select appropriate clustering and algorithms for various applications,		
M4.01	Association Rule Mining: Concepts- support and confidence	1	Apply
M4.02	FP-Growth Algorithm	3	Apply
M4.03	Cluster Analysis-Introduction, Concepts	1	Understanding
M4.04	Categorization and Clustering Method.	2	Understanding
M4.05	Partitioning Method: K-Means	2	Apply
M4.06	Hierarchical Clustering method: Agglomerative hierarchical clustering	2	Understanding
M4.07	Hierarchical Clustering method: BIRCH.	2	Understanding
	Series Test – II	1	

Contents:

Association Rule Mining: Concepts- support and confidence, FP-Growth Algorithm-
Cluster Analysis-Introduction, Concepts, Types of Data in cluster analysis, Categorization and Clustering Method.

Partitioning Method: K-Means and K-Medoid Clustering.

Hierarchical Clustering method: Agglomerative hierarchical clustering, BIRCH.

Text / Reference

T/R	Book Title/Author
T1	Dunham M H, “Data Mining: Introductory and Advanced Topics”, Pearson Education, New Delhi, 2003.
T2	Jaiwei Han and Micheline Kamber, “Data Mining Concepts and Techniques”, Elsevier, 2006.

Online Resources

Sl.No	Website Link
1	https://www.investopedia.com/terms/d/datamining.asp
2	https://www.techtarget.com/
3	https://www.javatpoint.com/data-mining
4	https://www.electronicshub.org/artificial-neural-networks-ann/
5	https://www.tutorialspoint.com/data_mining/
6	https://towardsdatascience.com/the-fp-growth-algorithm